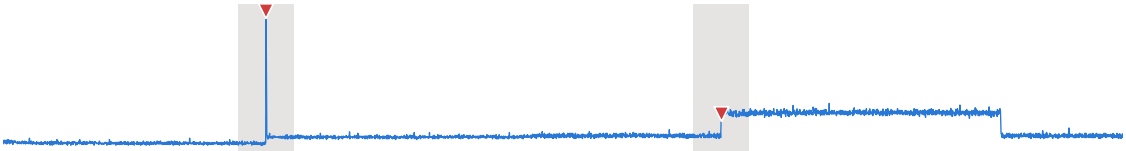
Sentinel

A live anomaly detector that learns what is **rare** for each signal, and explains every alarm in one sentence.



AWS database CPU, streamed point by point: a spike and a later level shift, both caught (▾). Grey bands: labelled anomaly windows.

0.594
event-F1 on all 58 NAB series

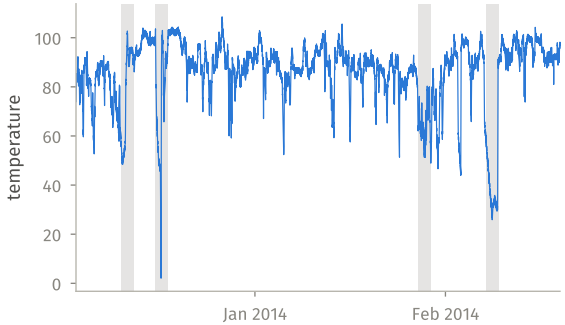
2.3×
the best standard baseline

61 µs
per point, on one CPU core

0
false alarms on clean signals

if/elif/else Aarohi Gattewar · Phalak Mehta · Harshit Singh

Catch real failures early, without crying wolf



A temperature sensor inside an industrial machine, ending in a known system failure. Grey: the 4 labelled anomaly windows.

1. **Decide live**
   Each value arrives once. Decide now, never peek ahead.
2. **Fit every signal**
   CPU, ad prices, taxis, tweets: no single threshold fits all.
3. **Few alarms, each explained**
   False alarms get ignored. Missed ones become outages.

How the challenge scores it: event-F1 over labelled anomaly windows, precision, recall, detection delay and runtime per point. Each alarm needs a score, a NORMAL/ANOMALOUS label and evidence.

A textbook 3σ rule raises 2,403 alarms for 116 real anomalies

Numenta Anomaly Benchmark

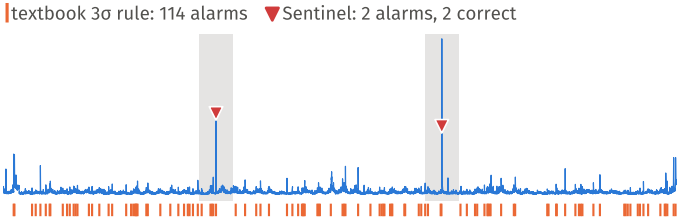
	series	points	windows
Servers (AWS)	17	67,740	30
Tweets	10	158,631	33
Known failures	7	69,561	19
Road traffic	7	15,664	14
Ads (hourly)	6	9,610	14
Artificial	11	44,352	6
Total	**58**	**365,558**	**116**

What the EDA showed

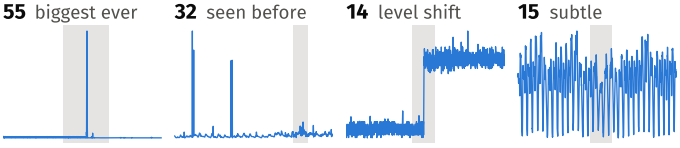
47 of 58 series have heavy tails.

18 have a daily rhythm.

Preprocessing: stream in time order, no filling or rescaling.

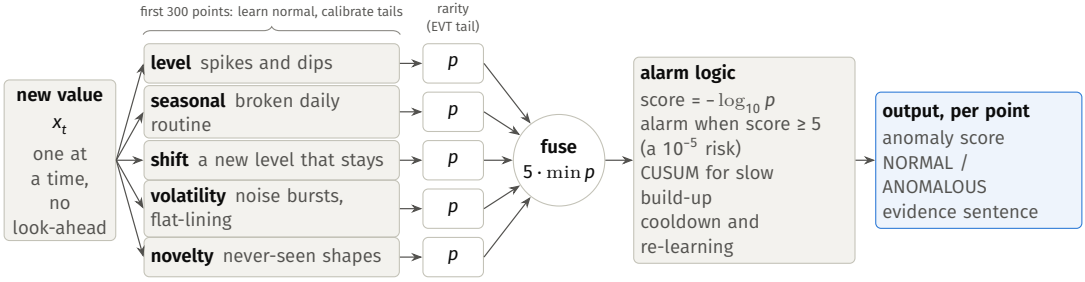


Tweets about Facebook: 2 labelled anomalies. Over all 58 series the 3σ rule raises 2,403 alarms; Sentinel raises 106.



All 116 windows by type. “Seen before”: an equally big spike came earlier, unlabelled.

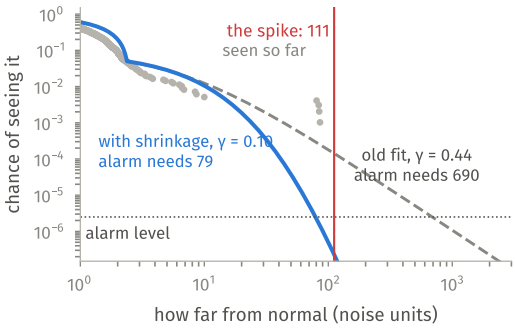
One pass per point: experts, calibrated rarity, explained alarm



strictly causal | O(1) state per expert | 61 µs per point | one configuration, 58 series | CSV + JSON + live demo

Features and windows. Level scores the current point; seasonal looks one day back; shift and volatility compare a 10-point with a 200-point average; novelty matches the last 8 points against the previous 2,000 (a streaming matrix profile). Every point is scored *before* the experts learn from it.

The threshold is a probability, not a number



One real spike in an hourly ad price, 111 noise units above normal. The old fit believed such values were common and would only alarm at 690. With shrinkage: 79, so it is caught.

Calibrated rarity, per signal, online

Extreme scores get an extreme-value tail, refitted in O(1):

$$p = \Pr(S > s) \approx \frac{n_t}{n}\left(1 + \gamma\,\frac{s - t}{\sigma}\right)^{-1/\gamma}$$

Alarm when $5 \cdot \min_e p_e < 10^{-5}$, on every signal.

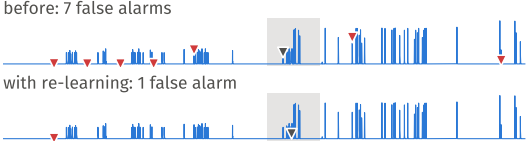
New: shape shrinkage

$\gamma \leftarrow \gamma \cdot \frac{n_t}{n_t + 200}$: trust a heavy tail only with evidence.

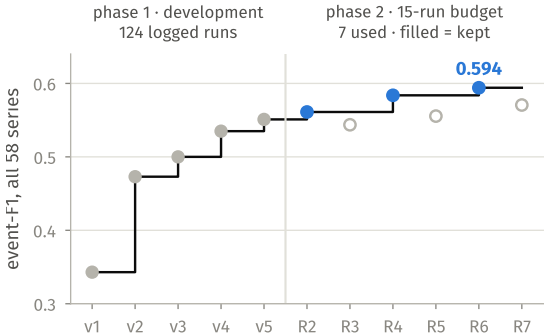
Biggest-ever spikes caught: 31 → 37.

New: re-learning

Still alarming after the cooldown? The normal has changed, so re-learn it.



15-run budget: one written hypothesis per run



Phase 1: v2 fixed tail learning, v3 added evidence accumulation, v4 the novelty expert, v5 set the risk and dropped a "jump" expert after an ablation.

run	change	F1	
1	reproduce v5 (reference)	0.551	
2	tail shrinkage, n_0 = 100	0.561	**kept**
3	stronger shrinkage, n_0 = 400	0.544	dropped
4	+ re-learning	0.584	**kept**
5	warm-up 300 → 200	0.556	dropped
6	shrinkage n_0 = 200	**0.594**	**kept**
7	shrinkage n_0 = 400	0.571	dropped

Search strategy

Analyse first: ideas come from saved outputs, at no cost.

One rule: keep a change only if F1 rises and more categories gain than lose.

Stop early: 7 ideas rejected from the data alone; 8 runs left in reserve.

Evaluation: event-level scoring, checked on unseen categories



Precision: share of alarms inside a window.
Recall: share of windows with at least one alarm.
Delay: first alarm minus the labelled anomaly point.

Validation

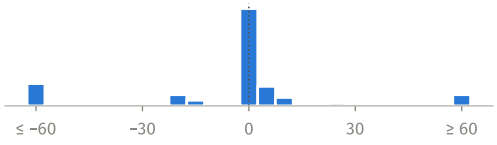
One configuration, all 58 series, strictly causal.

Leave one category out across all runs: **0.594** held-out.

The 5 anomaly-free series must stay silent: **0 alarms**.

No point adjustment: with it, random alarms on 1% of points score **0.92**, above our 0.82.

category	windows	P	R	F1
Server metrics (AWS)	30	0.67	0.47	0.55
Tweet volume	33	0.57	0.76	0.65
Known failures	19	0.52	0.63	0.57
Road traffic	14	0.80	0.29	0.42
Online ads (hourly)	14	1.00	0.36	0.53
Artificial	6	1.00	0.83	0.91
All	**116**	**0.63**	**0.56**	**0.594**



Detection delay: the median first alarm lands **0 steps** after the labelled anomaly point; 66% within ±10 steps.

0.594 event-F1: 2.3× the best baseline, at 61 µs per point

0.594
event-F1, all 58 series

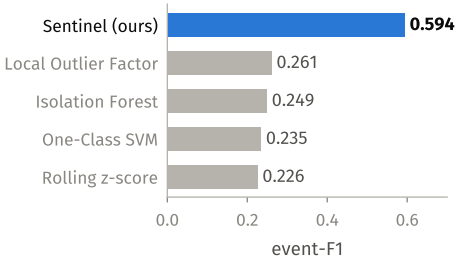
65 / 116
anomaly windows found

61 µs
per point (p50); 121 µs at p99

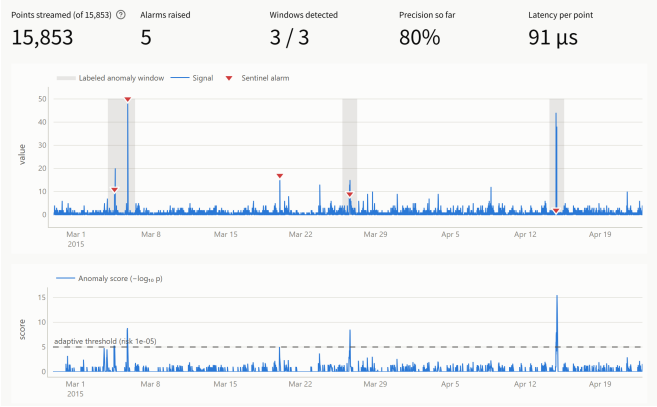
0
alarms on clean signals

method	F1	P	R	false alarms	µs/pt
Sentinel (ours)	**0.594**	**0.63**	0.56	**39**	61
Local Outlier Factor	0.261	0.17	0.62	455	216
Isolation Forest	0.249	0.18	0.42	261	49
One-Class SVM	0.235	0.14	0.68	594	9
Rolling z-score	0.226	0.15	0.47	351	0.5

Same data, same scoring. Each baseline got its best possible threshold, chosen with the labels; Sentinel uses one fixed configuration. The baselines reach their recall only by raising 7 to 15 times more false alarms.



Every alarm arrives with its evidence



Our Streamlit demo replaying tweets about CVS as a live stream: 3 of 3 anomaly windows found; alarms (▾) on the signal; the score against the adaptive threshold.

LEVEL · 5 Mar 19:57 · score 8.5

Value 50.00 is 99.4 scales above the learned baseline 0.32 (normal band −1.18 to 1.82).

SHIFT · 26 Mar 14:07 · score 5.1

Sustained level shift: recent mean 6.40 vs baseline 0.43 (8.2 scales above).

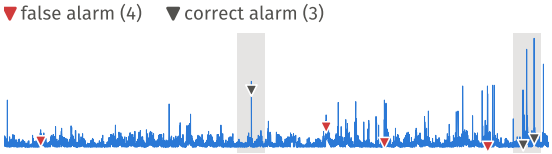
NOVELTY · 14 Apr 21:02 · score 6.0

The last 8 points form a pattern not seen before: the closest past match (1,750 steps ago) differs by 11.40 per point.

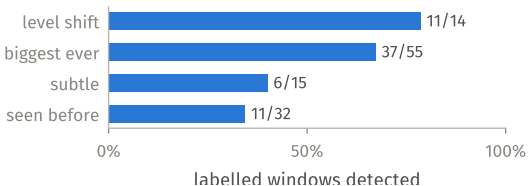
Same alarm in predictions.csv

```
2015-03-05 19:57:53 · 50.0 · score 8.525 · ANOMALOUS · level · "Value 50.00 is 99.4 …"
```

Where it still fails, and what comes next



Tweets about IBM. Most of our 39 false alarms are big bursts nobody labelled.



“Seen before” spikes stay hard: rarity alone cannot tell them apart.

Known limits

Seen-before spikes: 11 of 32 caught.

Short hourly series: Ad Exchange recall 0.36.

First 300 points only learn: 3 windows start there.

Next

Operator feedback: “not an anomaly” goes into the tail, so the pattern stops alarming.

Weekly and holiday rhythm for demand signals.

Signals from one machine together: CPU, disk and network.

Sentinel finds 65 of 116 anomalies with 63% precision, live at 61 µs per point. Every alarm is held to the same 1-in-100,000 risk and explained in one sentence.